Fictional enzyme X deficiency

Alex Example, 34 · tiredness and pale skin patches · an invented case

Worked example

Alex Example

TIME COURSE 1 from birth — 2 over years — 3 at the visit — 4 after the assay

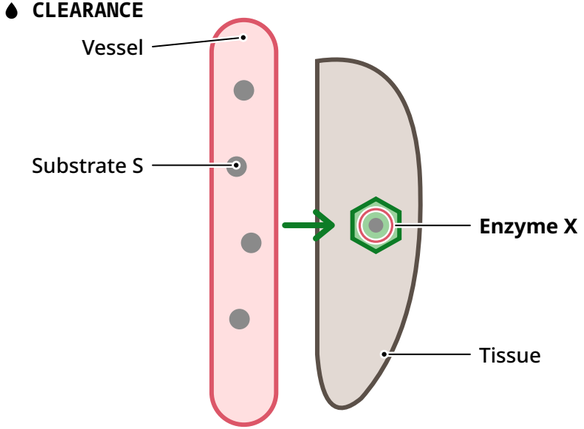
1 CAUSE · from birth

- Two faulty copies of gene XA, inherited from both parents
- Enzyme X normally clears substrate S from the blood

4 TREATMENT · after the assay

- Weekly enzyme X replacement given by infusion
- Low substrate S diet to slow the build-up

CLEARANCE



Drawn for this example

2 MECHANISM · over years

- Enzyme X activity falls below 10 % of normal
- Substrate S builds up in skin and muscle over years

KEY

- Risk factor
- Mechanism
- Normal anatomy or physiology
- Sign or symptom
- Investigation
- Diagnosis
- Treatment
- 1 Sequence number
- Causal link
- Required structure or function
- Evidence against diagnosis
- Blocked by treatment
- Pitfall: a common trap
- High-yield: a key concept
- ↑ ↓ Raised / reduced result
- Band: reference interval; dot: result; › beyond scale

3 FINDINGS AND DIAGNOSIS · at the visit

- Too much substrate S in the diet (ruled out)
- Enzyme X assay in blood cells: 6 % of normal activity
- **Fictional enzyme X deficiency, confirmed by the assay**
- Blood substrate S 48↑ µmol/L (normal 2–10)
- Tiredness after light exercise and pale skin patches